

A screened Rydberg formula for the successive ionization energies of all atoms and ions

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Abstract

We present a single closed-form expression, the *screened Rydberg formula*, for the successive ionization energy $IE(Z, N)$ of any atom or ion. It needs only the nuclear charge Z and the ground electron configuration. The formula keeps the hydrogenic form $Ry Z_{\text{eff}}^2/n^2$. Its effective charge is split into two parts. The first is an exact, parameter-free first-order screening constant $\sigma_1 = -n^2\Delta E_1$ from the $1/Z$ perturbation expansion; it is a rational number for every configuration, and we tabulate it for $N = 1$ –110. The second is a bounded, fitted higher-order remainder that depends on the ion charge in an Edlén-type form. A saturation construction guarantees $Z - N + 1 \leq Z_{\text{eff}} \leq Z$ for any parameter values (given $0 \leq \sigma_1 \leq N - 1$, which holds for every configuration we use) and keeps the exact Z^2 and Z coefficients of the $1/Z$ series. One-electron Dirac, recoil, finite-nuclear-size and QED corrections are added without fitted parameters. The final model has 33 fitted global parameters and no per-element or per-ion parameters. On all 5847 successive ionization energies in the NIST Atomic Spectra Database ($Z = 1$ –110), its mean absolute percentage error (MAPE) is 1.87 % (median 0.94 %). That is 7.52 % for neutral-atom first ionization energies and 0.0012 % for hydrogen-like ions; the H-like figure is consistent with, but not independent of, the QED theory behind the NIST values. The model was chosen by a pre-registered score that never used the blind extrapolation test (its interpolation splits do contain heavy elements). In a single blind extrapolation (fit $Z \leq 54$, predict $Z = 55$ –110), it gives 6.60 % MAPE (median 1.62 %). The same test gives 11.6 % for Slater's rules, 13.6 % for a published parameter-free screened-hydrogenic model scored on the same rows, and 57–178 % for earlier fitted variants without the bound. Extrapolation to heavy neutral atoms remains the weak point (21.8 % MAPE on their first ionization energies).

Keywords: ionization energy; screening constants; $1/Z$ expansion; screened hydrogenic model; isoelectronic sequences; Slater's rules; blind validation.

1. Introduction

The Bohr energy $Ry Z^2/n^2$ is exact only for a non-relativistic one-electron ion. For every other atom and ion the ionization energy (IE) is controlled by electron–electron repulsion, which has no closed-form solution for $N \geq 2$. Two broad routes exist.

Screening rules. Slater [Slater1930] replaced Z by an effective charge $Z - \sigma$, with σ given by simple counting rules fitted by hand to atomic data. Clementi and Raimondi [ClementiRaimondi1963; ClementiRaimondiReinhardt1967] obtained σ from optimised self-consistent-field orbital exponents of neutral atoms. Both are designed for orbitals of neutral atoms, not for the full set of successive IEs. Scored as IE formulas on all NIST ions with a total-energy difference, Slater's rules give 11.8 % MAPE and Clementi–Raimondi 120 % (§4).

Perturbation theory in $1/Z$. Layzer [Layzer1959; LayzerEtAl1964; Layzer1967] showed that the non-relativistic energy of a fixed configuration is an asymptotic series $E = Z^2 E_0 + Z E_1 + E_2 + \dots$. E_0 is hydrogenic and E_1 is a rational combination of hydrogenic Slater integrals, which fixes the exact $Z \rightarrow \infty$ limit of the screening constant. Higher orders need continuum sums [ScherrKnight1963; DalgarnoStewart1958]. Along isoelectronic sequences, Edlén [Edlen1964] systematised the smooth variation of screening with ion charge, using expansions in $1/(\zeta + s)$. Z -expansion codes with relativistic corrections were developed for selected sequences [Safronova1993].

Screened hydrogenic models (SHMs). Starting with Mayer [Mayer1947] and More [More1982], plasma-physics codes use screened hydrogenic energies with tabulated or fitted screening constants: l -splitting [Faussurier1997; Faussurier2008], analytical potentials [Martel1998; Rubiano2002], relativistic constants fitted by genetic algorithm [Mendoza2011], and a parameter-free, self-consistent (Z, N) -dependent screening [Kregar1984; Kregar1985; DiRocco1992; Pomarico2005; LanziniDiRocco2015; DiRoccoLanzini2016]. Average-atom and kinetics codes value these for speed and coverage of all ions [Rozsnyai1972; Chung2005; Crilly2023].

Ab initio methods. Koopmans' theorem [Koopmans1934], Δ SCF Hartree–Fock and Kohn–Sham DFT [KohnSham1965; Kotochigova1997], correlated non-relativistic energies [Chakravorty1993] and Dirac–Fock total energies for all ground configurations up to $Z = 118$ [Rodrigues2004] give more accurate numbers. They are numerical procedures, not formulas.

The gap. We found no single closed-form expression that (i) covers every ion of every element from the configuration alone, (ii) reproduces the exact large- Z behaviour of the $1/Z$ expansion, (iii) reports every fitted parameter, and (iv) is validated on all NIST successive IEs with held-out tests fixed in advance, including a blind extrapolation to heavier elements. The SHM papers we read report accuracy on their fitting data. For example, about 88 % of the database of [Mendoza2011] lies within ± 10 %, and [Pomarico2005] reports selected sequences. This paper tries to fill that gap. It does not claim a new law of atomic physics; §5.1 states what is rediscovered.

Contributions.

1. A table of the exact first-order screening constants σ_1 for every NIST ground configuration, $N = 1$ –110, as a parameter-free, $Z \rightarrow \infty$ -exact counterpart to Slater's σ (§2.1, §4.6).
2. The screened Rydberg formula: σ_1 plus a bounded Edlén-type remainder with 33 global parameters (§2.2–2.4).
3. A validation protocol with a pre-registered selection score and a single blind extrapolation test, applied identically to the final model, its ancestors and published baselines (§3, §4).
4. A head-to-head on the same 5847 rows against a published parameter-free SHM, which we re-implemented from its definitions (§4.5).

2. Theory

Atomic units are used in §2.1, with energies converted to eV via $Ry = 13.6057$ eV. Z is the nuclear charge, N the number of electrons before ionization, and $Z_a = Z - N + 1$ the charge seen by the departing electron. (n, l) is the removed subshell and k its occupancy.

2.1 Exact first-order screening from the $1/Z$ expansion

Scaling $r \rightarrow \rho/Z$ turns the non-relativistic Hamiltonian into $H = Z^2[H_0 + Z^{-1}V]$, with H_0 hydrogenic and $V = \sum \rho_{ij}$. Rayleigh–Schrödinger perturbation theory in $1/Z$ then gives, for a fixed configuration and term, $E(Z, N) = Z^2 E_0 + Z E_1 + E_2 + \dots$, and the ionization energy is

$$IE(Z, N) = Z^2 \Delta E_0 + Z \Delta E_1 + \Delta E_2 + \dots, \quad \Delta E_k = E_k(N-1) - E_k(N).$$

- **Zeroth order.** $\Delta E_0 = 1/(2n^2)$, which is exactly Bohr's formula.
- **First order.** $E_1 = \langle V \rangle$ over the $Z = 1$ hydrogenic state. Every hydrogenic radial integral F^k and G^k is an exact rational number, because each product $P_a P_c$ is a polynomial times $e^{-\beta r}$ with rational β . Examples: $F^0(1s, 1s) = 5/8$, $F^0(1s, 2s) = 17/81$, $G^0(1s, 2s) = 16/729$. We evaluate them in exact rational arithmetic and combine them with exact angular algebra.
- **Open shells.** For open subshells, E_1 is evaluated for the Hund ground term by projecting onto the highest-weight (S, L) subspace. Where hydrogenic degeneracy couples configurations (the Layzer complex, e.g. $1s^2 2s^2$ with $1s^2 2p^2$), V is diagonalised within the complex. For two or more open subshells, a high-spin coupled term is used; this is the one approximation in σ_1 .

Completing the square in the first two terms defines the first-order screening constant

$$IE \simeq \Delta E_0 (Z - \sigma_1)^2, \quad \sigma_1 = -\frac{\Delta E_1}{2\Delta E_0} = -n^2 \Delta E_1.$$

σ_1 is Layzer's $Z \rightarrow \infty$ screening constant. It has no adjustable parameter and depends only on the configuration, not on Z . Two checks against known values:

- **He-like:** $E_1(1s^2) = 5/8$, so $\sigma_1(\text{He}) = 0.6250$ (Slater: 0.30).
- **Li-like:** $E_1(1s^2 2s) = 5965/5832 = 1.0228052 = 5/8 + 2 \cdot 17/81 - 16/729$, so $\Delta E_1 = -0.397805$ and $\sigma_1 = 1.5912$.
- **Be-like:** the complex value $E_1 = 1.5592742$ agrees with Layzer's.

The exact series fails for near-neutral atoms (§4.6). The expansion parameter is effectively N/Z . For a neutral atom, $Z - \sigma$ is only 1–3, so a 10 % error in σ becomes a 100–1000 % error in IE . A closed form for the higher orders does not exist: E_2 already requires sums over the hydrogenic continuum. For example, the completed-square guess $\Delta E_2 = \Delta E_1^2 / (4\Delta E_0)$ is 24–26 % too large for He- and Li-like ions. This is why the remainder in §2.2 has to be fitted.

2.2 The screened Rydberg form with a bounded remainder

The final model ("pa_hier_rel") is

$$IE(Z, N) = \mu(Z) \left\{ \text{Ry} \frac{Z_{\text{eff}}^2}{n^2} D_{n,j}(Z_{\text{eff}}) \left[1 + r_c \frac{(Z\alpha)^2}{n} \left(\frac{Z_{\text{eff}}}{Z_a} - 1 \right) \right] + \text{Ry} \frac{x_l K_l(k)}{n^2} \right\} - [\Delta E_{\text{QED}} + \Delta E_{\text{FNS}}]_{Z,n}$$

where the last term applies only to the removal of an ns electron with $n \leq 2$, and

$$Z_{\text{eff}} = Z - \sigma_1(\mathcal{C}) - D, \quad D = \frac{T}{Z_a + \kappa + |T|/h}, \quad h = \begin{cases} (N-1) - \sigma_1 & T \geq 0 \\ \sigma_1 & T < 0 \end{cases}, \quad T = \sum_g \tau_g \nu_g + \sum_c$$

The terms are:

- **ν_g** counts the other electrons in five screening groups (same subshell; inner, i.e. same- n s and the $(n-1)$ shell, for s/p targets; deeper core for s/p targets; all $n' \leq n$ for d/f targets; outer $n' > n$). Each group has a coefficient τ_g .
- **ν_c and $\delta\tau_c$** are finer screening classes, with deviations shrunk toward their group value by a ridge penalty (10^{-4} per row).
- **$K_l(k) = [P(k) - P(k-1)] - 2l(k-1)/(4l+1)$** is the change in the number of parallel-spin pairs relative to a statistical average (Hund kink). For p electrons it is 0, 0.6, 1.2, -1.2, -0.6, 0 for $k = 1 \dots 6$. Its amplitude x_l is fitted per l .
- **$\mu(Z)$** is the reduced-mass factor.

The saturation form of D has two properties that hold for any parameter values.

1. **The bound $Z_a \leq Z_{\text{eff}} \leq Z$.** Write $D = \text{sign}(T) \cdot h \cdot u / (1+u)$ with $u = |T|/[h(Z_a + \kappa)]$. Then $|D| < h$, so Z_{eff} can neither fall below the fully screened, non-penetrating limit Z_a nor exceed the bare charge Z . Since $h = (N-1) - \sigma_1$ or σ_1 , this requires $0 \leq \sigma_1 \leq N - 1$. That is a checked property of the configurations, not a proven one: it holds for all 7021 ground/Madelung configurations with $Z \leq 118$, with no violations, and the code warns if a user-supplied configuration breaks it.
2. **The exact asymptotics.** At fixed N and $Z_a \rightarrow \infty$, $D \rightarrow T/(Z_a + \kappa) = O(1/Z)$. The expansion of IE therefore begins $Ry[Z^2 - 2Z\sigma_1 + O(1)]/n^2$, and its Z^2 and Z coefficients are those of the exact $1/Z$ series. The fitted part only represents ΔE_2 and higher orders: relaxation, correlation and penetration at low ion charge.

2.3 Charge-dependent penetration as an Edlén-type term

The remainder's dependence on $1/(Z_a + \kappa)$ was found empirically (docs/semi_empirical.md). Along every isoelectronic sequence, the excess charge $p = Z_{\text{eff}} - Z_a$ grows with ion charge q as $p \propto -\tau/(Z_a + \kappa)$, with one κ common to all sequences. For the Na sequence (3s), for example, $p = 0.84, 1.15, 1.34, 1.46, 1.56$ for $q = 0-4$ and 2.11 at $q = 20$. Introducing this term reduced the error of the purely fitted model from about 9 % to about 2 % MAPE. This is an independent rediscovery of the isoelectronic regularity that Edlén formalised [Edlen1964]. It is also consistent with Layzer's theory, in which the screening constant is $\sigma_0 + \sigma_1'/Z + \dots$. We do not present it as a new law. What is specific here is narrower: one universal denominator across all sequences, combined with exact σ_1 and with the saturation bound of §2.2.

2.4 Relativistic, recoil, finite-size and QED corrections

- **$D_{\{n,j\}}(\zeta)$** is the exact ratio of the point-nucleus Dirac binding energy to the Schrödinger energy for charge ζ , with j assigned by jj filling: $l - 1/2$ while $k \leq 2l$, otherwise $l + 1/2$.
- **The bracket $1 + r_c(Z\alpha)^2(Z_{\text{eff}}/Z_a - 1)/n$** is a Fermi-Segrè-type correction for the penetration of the valence electron into the region of full nuclear charge. Its five coefficients r_c (classes s, $p_{1/2}$, $p_{3/2}$, d, f) are fitted.
- **One-electron terms.** For one-electron ions we use the Dirac energy with a numerically solved finite-nuclear-size (FNS) shift, Barker-Glover recoil, the Uehling vacuum polarisation computed over the Dirac 1s density, and the one-loop self-energy function $F_{\text{SE}}(Z\alpha)$ from the all-order tabulation of Yerokhin and Shabaev [YerokhinShabaev2015]. The 2×110 tabulated values are external theory inputs, not fitted parameters. In the many-electron formula, the QED and FNS shifts are applied only to 1s and 2s removal, scaled by $(Z_{\text{eff}}/Z)^2$.
- **No closed-form substitute for the table.** The closed-form low-order $Z\alpha$ expansion of F_{SE} matches the table at $Z = 1$ but diverges for $Z \gtrsim 15$. It gives 0.354 % MAPE on H-like ions, worse than no QED.

2.5 Density functional theory as a physics check

We also wrote a radial Kohn-Sham LSDA solver (Slater exchange + VWN5 correlation). It reproduces the NIST LDA reference total energies [Kotochigova1997] to about 10^{-6} hartree (e.g. Ne -128.233481). Because that reference uses the same functional, the agreement verifies the implementation only, not physical accuracy. ΔSCF ionization energies from this solver have no fitted parameters and serve as an independent, non-formula check (§4.2). DFT does not enter the formula.

2.6 Parameter count

The final model has 33 fitted global parameters:

block	count
group screening τ_g	5
saturation denominator κ	1
class deviations $\delta\tau_c$ (ridge-shrunk; counted in full)	19

block	count
relativistic penetration r_c	5
Hund amplitudes x_l (p, d, f)	3

There are no per-element or per-ion parameters, and the prediction code never reads a NIST ionization energy. Fitted values are in Table A1 (Appendix) and `results/uni_final_params.json`.

3. Data and validation protocol

3.1 Data

The reference data are the 5847 successive ionization energies of the NIST Atomic Spectra Database [NISTASD] for $Z = 1-110$, all charge states, with NIST ground configurations (`data/nist_ie.csv`). The database flags each value as experimental (311 rows), semi-empirical (919) or theoretical (4617). Every metric below is computed over all 5847 rows unless stated otherwise. By status, the final model's all-data fit gives (`evaluate.py`, `status= strata`): experimental 4.63 % MAPE (median 2.86 %, $n = 311$), semi-empirical 2.20 % (median 0.83 %, $n = 919$), theoretical 1.62 % (median 0.92 %, $n = 4617$). The experimental rows are mostly neutral atoms and low-charge ions (98 neutral, 193 of 311 with charge ≤ 2 , median charge 2), the hardest regime for the formula, so their higher error mirrors the neutral-atom weakness rather than a disagreement with experiment specifically. **[TODO: insert the ASD version and access date.]**

Subsets used below:

- 108 neutral-atom first IEs;
- 110 hydrogen-like ions;
- 63 "rearranged" ions, whose ground configuration changes by more than one electron on ionization (e.g. $V\ 3d^3 4s^2 \rightarrow V^+ 3d^4$).

The configuration of each ion is an input. Inside the table it is the NIST ground configuration; beyond $Z = 110$ the Madelung order is used.

The error measure is $MAPE = (100/M) \sum |IE_{pred} - IE_{NIST}| / IE_{NIST}$, together with the median absolute percentage error. All scores come from one shared scorer (`evaluate.py`). Fits minimise squared log-ratios $\ln(IE_{pred}/IE_{NIST})$, so a 4 eV and a 100 keV ionization energy carry equal relative weight.

3.2 Held-out splits

Every fitted model is refit on the training rows of each split with its own fitting routine:

split	training rows	test / validation rows	n test rows	role
V1	$Z \leq 36$	$37 \leq Z \leq 54$	819	selection
V2	$Z \leq 44$	$45 \leq Z \leq 54$	495	selection
S1	$Z \bmod 5 \neq 0$	$Z \bmod 5 = 0$ (unseen elements)	1188	selection
S3	$N \bmod 6 \neq 0$	$N \bmod 6 = 0$ (unseen isoelectronic sequences)	928	selection
S2	$Z \leq 54$	$Z \geq 55$	4362	blind , reported once

Selection score. The score is the mean MAPE over V1, V2, S1 and S3. It was fixed before the final round of model development and was the only quantity used to choose the final model. S2 (fit $Z \leq 54 \rightarrow Z \geq 55$) was never used for any choice: it was computed once per candidate, after its specification was frozen. The selection splits themselves are not free of heavy elements, however. As pre-registered, S1 and S3 contain $Z \geq 55$ rows in both training and test sets (913 of the 1188 S1 test rows and 703 of the 928 S3 test rows, about 76 %), so

heavy-atom *interpolation* accuracy did inform the design choices; only heavy-atom *extrapolation* was kept blind. As a robustness check (reporting only, after the audit), we recomputed the selection score with S1 and S3 restricted to $Z \leq 54$ in both training and test. The winner is unchanged: pa_hier_rel 1.844, then the 28-parameter variant without the relativistic term 1.989, the 9-parameter bounded variant 2.094 and the best Push B candidate 2.118. This checks the ranking of the frozen candidates, not the exploration path that produced them. The final model's numbers were reproduced by an independent validation script, which matched the developing agent's own run with a difference of 0.0 in every split.

3.3 History of the protocol (disclosure)

The protocol above was the third. In earlier stages of this project:

1. A purely fitted model reported S2 = 19 % using a bound on core screening chosen after seeing S2. Without that bound, the blind value is 170 %.
2. Later, an intermediate unified model was chosen on a score that included S2.

Both are reported here only as non-blind history (§4.3). Other disclosures:

- The developers of the final round knew which heavy near-neutral atoms the previous model failed on, and that an 8-parameter model extrapolated well. This motivated the direction of the search, though not the choice, which was made on the selection score.
- About 57 variants were explored in that round (43 logged by one agent, about 14 by the other). The minimum selection score over them is therefore optimistic.
- The screening classes and the Hund term were designed earlier on all rows, including $Z \geq 55$. This is structural leakage that we cannot remove retrospectively.

4. Results

4.1 Baselines versus the final model

Table 1. All values are MAPE in %. "All", "neutral" and "H-like" come from all-data fits. V1, V2, S1, S3 and S2 are held-out values after refitting. For parameter-free models nothing is fitted, so the split columns are simply the error on those rows. Sources: results/model_comparison.md, results/uni_validation.md.

model	fitted params	all (median)	neutral 1st IE	H-like	V1	V2	S1	S3	selection score	blind S2 (median / neutral)
Bohr Ry Z^2/n^2	0	1363 (172)	$2.3 \cdot 10^4$	6.00	1050	1045	1317	1399	1203	1517
Slater 1930, total-energy difference	0	11.8 (7.48)	51.8	6.00	13.4	14.4	11.6	12.4	12.9	11.6 (8.07 / 50.7)
Clementi–Raimondi, total-energy difference	0	120 (18.7)	2281	6.00	87.1	77.3	116	119	99.9	134
Exact 1/Z series, completed square + rel/QED ("zexp")	0	69.8 (7.20)	1105	0.0012	53.4	54.0	68.3	74.3	62.5	77.8 (– / 1772)
GSHM final (purely fitted, blind)	32	1.57 (0.74)	7.61	0.19	34.1	1.65	1.59	1.70	9.77	170 (3.51 / 7393)
exact σ_1 + linear remainder ("u35")	35	1.48 (0.73)	6.43	0.0012	26.4	1.48	1.52	1.65	7.76	57.2 (2.00 / 3831)
pocket formula (hand- calculable)	8	4.68 (3.04)	16.8	1.16	$4.4 \cdot 10^6$ (fit diverges)	2.55	4.62	5.24	–	11.3 (2.04 / 12.3)
lighter bounded variant ("pa_bound9")	9	2.90 (1.39)	12.0	0.0012	2.16	1.95	2.90	3.05	2.51	7.80 (– / 28.0)

model	fitted params	all (median)	neutral 1st IE	H-like	V1	V2	S1	S3	selection score	blind S2 (median / neutral)
Screened Rydberg formula (final, "pa_hier_rel")	33	1.87 (0.94)	7.52	0.0012	3.09	1.60	1.91	2.00	2.15	6.60 (1.62 / 21.8)

The final model has the lowest selection score of all candidates (2.15). The next were the other developer's choice, "pb_clip_pos" (30 parameters), at 2.37 and the 28-parameter hierarchical variant "pa_hier" at 2.37 (blind S2 13.9 % and 8.47 %, respectively). In-distribution, the bound costs a little accuracy: 1.87 % against 1.48 % for u35, and 7.52 % against 6.43 % on neutral atoms. In return, it removes the catastrophic extrapolation failures.

4.2 Errors by stratum

Table 2. All-data fit, MAPE in % (source: [results/lit_comparison.md](#) / [docs/literature.md](#) §2.2).

model	$N \leq 10$	$11 \leq N \leq 36$	$N \geq 37$	$Z \geq 55$	removed d	removed f
Slater 1930	4.89	9.38	16.8	11.6	16.2	12.9
pocket formula (8 p)	2.37	2.76	7.38	4.7	3.67	11.5
GSHM final (32 p)	0.698	1.16	2.31	1.48	0.982	2.67
u35 (35 p)	0.463	1.08	2.28	1.58	1.17	2.58
final (33 p)	0.439	1.43	2.87	1.92	1.52	3.51

- **Charge.** The error grows with the number of electrons and falls with ion charge. On ions of charge ≥ 3 the final model gives 1.61 %.
- **Where theory alone suffices.** For few-electron, highly charged ions, the parameter-free exact series is already good: median 0.7 % for $N \leq 10$, and 0.48 % MAPE for $N/Z \leq 0.2$ with two terms.
- **Neutral atoms, DFT check.** The parameter-free LSDA Δ SCF solver reaches 3.30 % on the 54 neutral first IEs with $Z \leq 54$, and 0.80 % on all 171 ions with $Z \leq 18$. It is better than every closed form on neutral atoms, but it is a numerical procedure and was run on only 207 rows.

Figures (existing): parity plot [results/figures/uni_parity.png](#); first IEs against Z [results/figures/uni_first_IE.png](#); successive IEs of selected elements [results/figures/uni_successive.png](#); residuals [results/figures/uni_residuals.png](#).

4.3 Blind extrapolation (S2)

With a fit on $Z \leq 54$ only, the final model predicts the 4362 ionization energies of $Z = 55$ –110 with 6.60 % MAPE and a 1.62 % median. The ancestors of the formula show what each ingredient contributes:

model	what it adds	blind S2 MAPE (%)	S2 neutral (%)
GSHM final (fitted screening, no exact σ_1)	–	170	7393
u35	exact σ_1 as the large-Z anchor, linear remainder	57.2	3831
final	+ saturation bound $Z_a \leq Z_{\text{eff}} \leq Z$, hierarchical shrinkage	6.60	21.8
Slater 1930 (0 p, nothing extrapolated)	–	11.6	50.7

- **What fixed extrapolation.** Exact σ_1 alone was not enough. The decisive step was confining the fitted part to a bounded remainder. In a 14-parameter ablation, the bound improved the selection score from 3.96 to 2.91; its blind S2 changed from 9.43 % to 5.53 %, and that S2 change was seen only after the choice.
- **Ions versus neutral atoms.** For ions ($Z > N$) the blind S2 MAPE is 6.4 %. For neutral atoms it is 21.8 %.

- **Per element.** Figure 1 shows the per-element median error on the S2 test rows.
- **Does the selection score predict extrapolation?** Figure 2 plots selection score against blind S2 for all fitted candidates. Candidates with a low selection score generally extrapolate well, but the relation is not monotone. A candidate with a worse selection score (pb_exp_pos, 3.73) has a lower blind S2 (6.03 %) than the final model. It was not promoted, because doing so would be selection on S2.

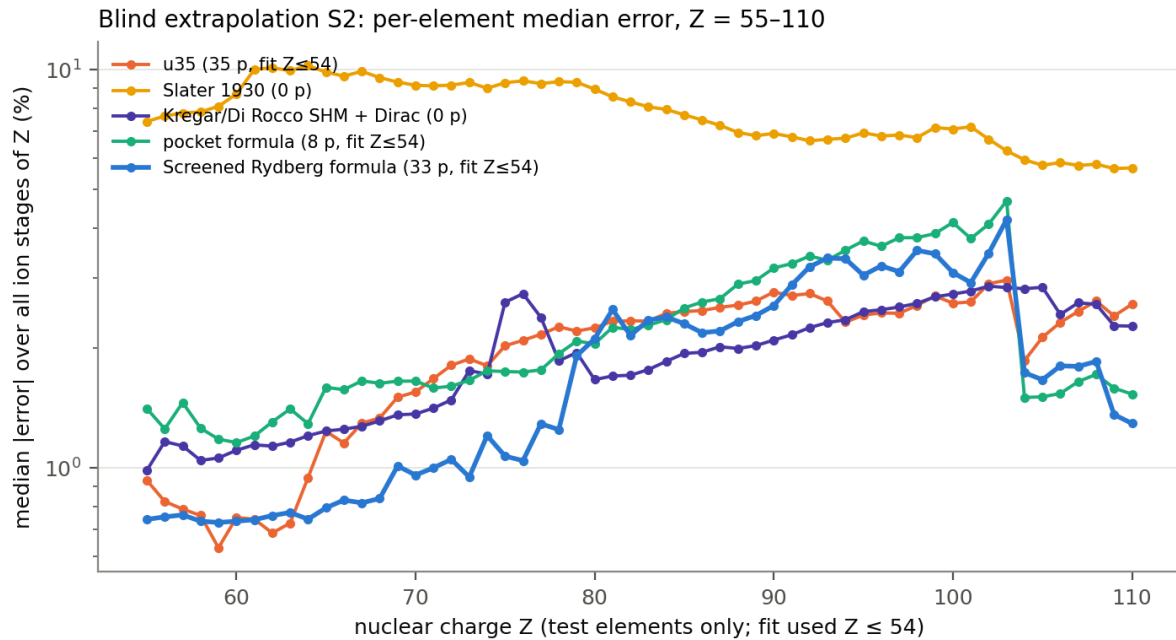


Figure 1. Blind extrapolation. Each point is the median absolute percentage error over all ion stages of one element $Z = 55-110$. The three fitted models are fitted on $Z \leq 54$ only; Slater's rules and the Kregar/Di Rocco SHM have no fitted parameters. Generated from refits with the project's validation code; the S2 MAPEs reproduce Table 1 (final 6.603 %, u35 57.233 %, pocket 11.334 %, Slater 11.638 %).

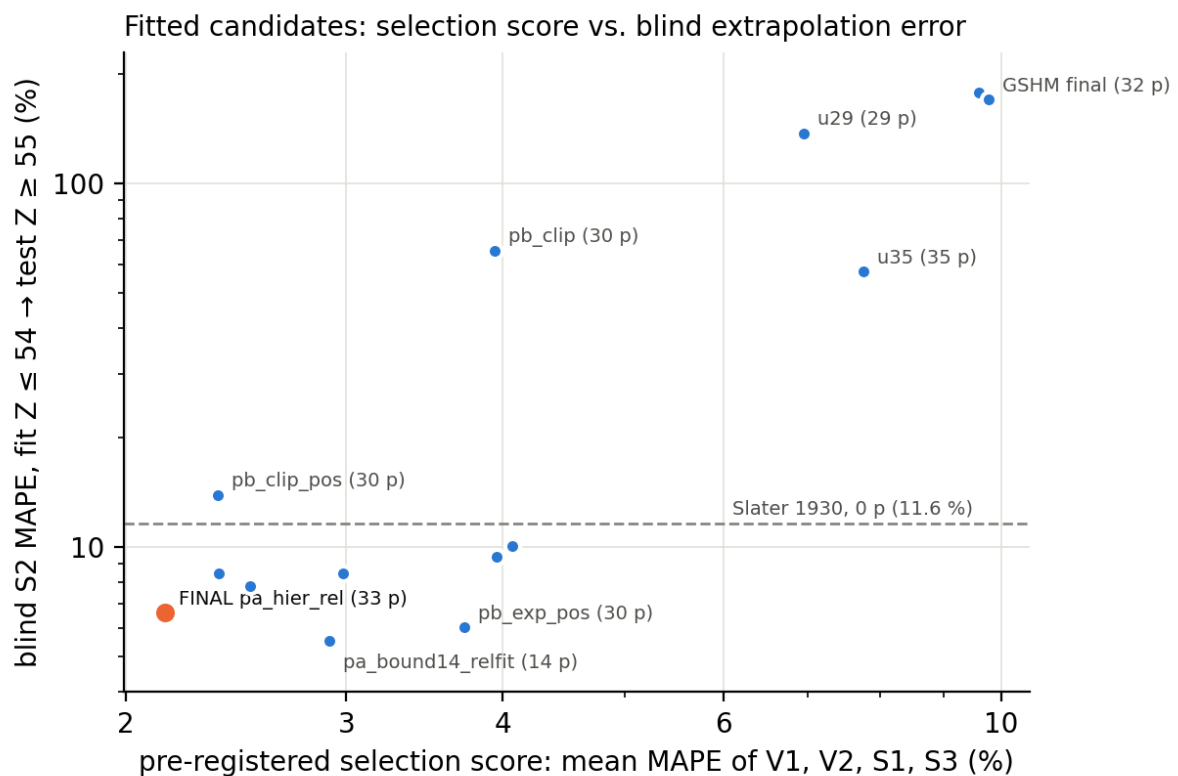


Figure 2. Selection score (mean of V1, V2, S1, S3; no S2 data, although S1 and S3 contain $Z \geq 55$ rows) against blind S2 MAPE for every fitted candidate in `results/model_comparison.csv`. The pocket formula is omitted because its V1 fit diverges. The dashed line is Slater's rules (0 parameters).

Failures of the blind fit, all reported here and none corrected, since correcting them now would be post hoc:

- **Heavy p-block neutrals.** Their first IEs are badly underestimated. Pb is predicted at 1.19 eV against 7.42 eV, Tl at 1.73 against 6.11, and Rn at 6.10 against 10.75.
- **Lr is negative.** Lr is predicted at -3.33 eV. The fitted r_c are negative, and the relativistic bracket is not bounded, so the Z_{eff} bound alone does not guarantee a positive IE once $Z\alpha$ is large.
- **f removal.** Third IEs of lanthanides and actinides (4f/5f removal) are about 2.0–2.5 times too high (mean 2.2, median 2.1, 23 rows; worst Er^{2+} at 55.65 eV against 22.7 eV), because the $Z \leq 54$ training set contains no f electrons.

4.4 Hydrogen-like ions: correction layers

Table 3. 110 H-like ions, $Z = 1\text{--}110$ (`docs/first_principles.md` §2.1).

layer	MAPE (%)	median (%)	max (%)
Bohr Ry Z^2	6.0	4.2	19.5
(a) Dirac, point nucleus	0.190	0.118	0.91
(b) + recoil + finite nuclear size	0.113	0.109	0.248
(c) + one-loop QED (Uehling computed; F_{SE} from [YerokhinShabaev2015])	0.00117	0.000154	0.0096
(d) as (c) but with the closed-form $Z\alpha$ expansion of F_{SE}	0.354	0.191	1.18

Caveat. The NIST H-like reference values are themselves computed from the same QED theory. Layer (c) therefore shows consistency with that source, not independent validation. The remaining $10^{-5}\text{--}10^{-4}$ residual at high Z is the omitted two-loop QED, nuclear-polarisation and recoil-QED terms.

4.5 Head-to-head with a published screened hydrogenic model

Of the published SHMs, only the parameter-free Kregar/Di Rocco model could be fully specified from articles we were able to read [Pomarico2005; DiRoccoLanzini2016]. We implemented it from its definitions:

- screening from hydrogenic densities with an exchange correction, iterated to self-consistency;
- $E = -\sum q_i Z_i^2 / 2n_i^2$;
- non-relativistic, Pauli and Dirac variants.

Two other cases:

- The constants of More [More1982] and Faussurier et al. [Faussurier1997] are in papers we could not access, and no verifiable reprint of the tables was found. We did not reconstruct them from memory.
- The constants of Mendoza et al. [Mendoza2011] were also not obtained.

Fidelity of our implementation.

- **Matches the published numbers:** same-shell $Z \rightarrow \infty$ screening constants reproduce every printed digit (e.g. 1s 0.3125, 2p 0.3492), and total energies agree with the published table within 0.71 %.
- **Differs:** cross-shell constants differ by 0.011 on average (at most 0.053). The original uses fitted closed forms whose coefficients we could not obtain [DiRocco1992]. Valence IEs of near-neutral ions come out 3–5 eV higher than the printed model values; Ar I, for example, is 18.96 eV here against 14.72 eV printed.

Our near-neutral numbers are therefore "the model as defined", not the authors' code.

Table 4. Same 5847 rows, same scorer, MAPE in % (`results/lit_comparison.md`). For the 0-parameter models the S2 column is the error on the $Z \geq 55$ rows. For the fitted models it is the blind S2 refit.

model	params	all (median)	neutral 1st IE	H-like	charge ≥ 3	S2 ($Z \geq 55$)
Kregar/Di Rocco SHM, non-relativistic	0	13.6 (2.34)	230	6	–	15.1
Kregar/Di Rocco SHM + Pauli	0	12.6 (1.79)	225	1.16	–	13.9
Kregar/Di Rocco SHM + Dirac	0	12.5 (1.71)	224	0.19	4.68	13.6
Slater 1930	0	11.8 (7.48)	51.8	6	10.2	11.6
pocket formula	8	4.68 (3.04)	16.8	1.16	–	11.3
final	33	1.87 (0.94)	7.52	0.0012	1.61	6.6

The 224 % neutral figure is for our implementation, built from the published definitions. It is not fully faithful for valence shells: it gives 18.96 eV for Ar I where the authors print 14.72 eV, and the authors' printed valence IEs are 3–5 eV lower than ours, so this figure probably overstates the error of the authors' own model on neutral atoms.

The parameter-free SHM works well for ionised species, with a median of 1.54 % for charge ≥ 3 . It fails near neutrality: 37 of its 5847 predictions are zero or negative, all for ions of charge 0–4 with an open d or f shell. The fair conclusion has three parts:

1. Where the $Z \rightarrow \infty$ limit dominates, parameter-free and exact- σ_1 models both work.
2. Near neutrality, every parameter-free hydrogenic model degrades. The fitted remainder is what makes the formula useful there, at the price of 8–33 parameters.
3. The bounded fitted formula extrapolates better (6.6 %) than the parameter-free SHM scores on the same $Z \geq 55$ rows (13.6 %). The unbounded fitted models extrapolate worse (57 % and 170 %).

A comparison with Dirac–Fock ionization energies from [Rodrigues2004] on the same rows has not yet been done. **[TODO]**

4.6 The exact first-order screening constants compared with Slater's

Table 5. σ_1 for neutral-atom ground configurations, compared with Slater's σ . Excerpt; the full table for $N = 1$ –110 is in `results/fp_zexp_coefficients.csv`, with values for all 5847 rows in `results/fp_zexp_rows_coefficients.csv`.

N	configuration	removed	E_1 (exact)	ΔE_1	σ_1 (ab initio)	σ (Slater)
2	$1s^2$	1s	5/8	–0.625000	0.6250	0.30
3	$1s^2 2s$	2s	5965/5832	–0.397805	1.5912	1.70
4	$1s^2 2s^2$	2s	586373/373248	–0.536469	2.1459	2.05
6	$1s^2 2s^2 2p^2$ (3P)	2p	10957679/3359232	–0.931338	3.7254	2.75
8	$1s^2 2s^2 2p^4$	2p	4754911/839808	–1.308370	5.2335	3.45
10	$1s^2 2s^2 2p^6$	2p	2455271/279936	–1.636495	6.5460	4.15
11	[Ne]3s	3s	9.635901	–0.865071	7.7856	8.80
18	[Ne]3s ² 3p ⁶	3p	17.980333	–1.407535	12.6678	11.25
19	[Ar]4s	4s	18.859971	–0.879639	14.0742	16.80
29	[Ar]3d ¹⁰ 4s	4s	39.317547	–1.347129	21.5541	25.30
36	[Ar]3d ¹⁰ 4s ² 4p ⁶	4p	50.145538	–1.667284	26.6765	27.75

Notes:

- E_1 is the Hund-term single-configuration value. For $N = 4$ (Be) the Layzer-complex value 1.5592742 is used in σ_1 . Decimal E_1 entries are exact rationals stored in the project cache.

- $\sigma_1 = -n^2\Delta E_1$.

Slater's rules describe the neutral atom; σ_1 is the exact screening of the bare-nucleus limit. The two disagree in a systematic way (Figure 3):

- **Inner electrons:** the hydrogenic same-shell screening is stronger than Slater's 0.35 per electron (Ne 2p: 6.55 against 4.15).
- **Valence electrons of heavy atoms:** σ_1 is much smaller than Slater's σ and than the screening implied by the measured IE (Na 3s: 7.79 against Slater 8.80 and an empirical 9.16; Cs 6s: 42.8 against Slater 52.8).

This difference is exactly what the fitted remainder D supplies near neutrality.

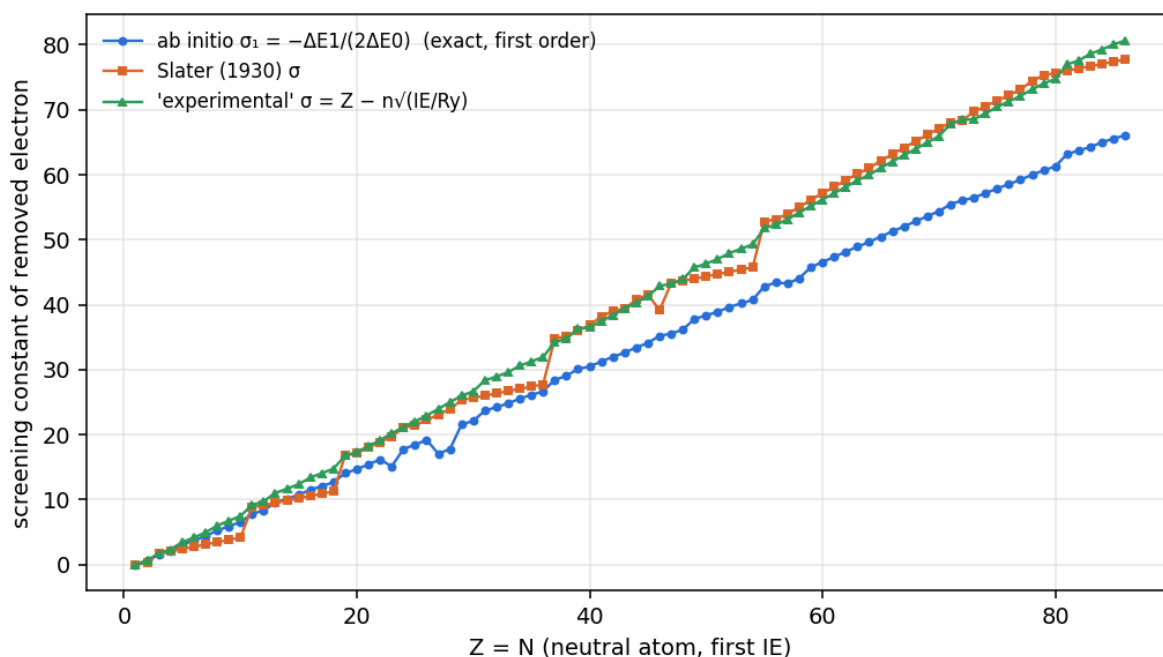


Figure 3. Exact first-order screening σ_1 against Slater's σ for neutral-atom configurations (existing project figure).

4.7 Worked example and coverage

For the first IE of oxygen ($1s^2 2s^2 2p^4 \rightarrow 2p^3$, NIST 13.618 eV):

1. $\sigma_1 = 5.23348$ and $T = 3.3496$.
2. $h = 1.76652$, so $D = 0.71285$ and $Z_{\text{eff}} = 2.05367$, which lies between $Z_a = 1$ and $Z = 8$.
3. The hydrogenic term is 14.3457 eV, or 14.3162 eV after relativity.
4. The Hund term is -0.8365 eV.
5. IE = 13.479 eV, an error of -1.02% .

The analogous 8-parameter pocket calculation gives 9.70 eV (-28.8%). For Mg^{2+} (NIST 80.144 eV), the formula gives 78.63 eV (-1.88%). Full arithmetic is in `docs/unified.md` §6.

Coverage test. The public API was evaluated for every $Z = 1-118$ and every $N = 1-Z$: 7021 values. All are finite and positive, with no crashes. There are 22 violations of the monotonicity $\text{IE}(Z, N-1) > \text{IE}(Z, N)$:

- 6 at Pt–Bi, caused by 4f/5s ordering;
- 16 at Rf–Ds, where the input ground configuration jumps between neighbouring ions (e.g. Rf $N = 68$ is $4f^{12}6s^2$, $N = 69$ is $4f^{14}5d^1$).

Above $Z = 110$ the predictions are qualitative. The 7p neutrals are too low: Og is predicted at 3.66 eV, implausibly below its lighter congener Rn (NIST 10.75 eV), and the cause is again the negative r_c . **[VERIFY: if published relativistic coupled-cluster values for Cn, Fl and Og are to be quoted, cite them from a checked source; the project's approximate values are from memory and must not be printed as literature.]**

5. Discussion

5.1 What is new and what is rediscovered

Rediscovered, and credited as such:

- the $1/Z$ expansion and its exact first-order term (σ_1 is Layzer's $Z \rightarrow \infty$ screening constant [Layzer1959], computed with textbook Slater-integral algebra [Cowan1981; FroeseFischer1997]);
- the screened hydrogenic form [Slater1930; Mayer1947; More1982], and self-consistent (Z , N)-dependent screening [Kregar1984; Pomarico2005];
- the $1/(Z_a + \kappa)$ remainder, which is Edlén-type isoelectronic behaviour [Edlen1964];
- the one-electron Dirac, recoil, finite-size and QED corrections [YerokhinShabaev2015; NISTASD];
- LSDA Δ SCF [KohnSham1965; Kotochigova1997].

Plausibly new, and claimed with care:

1. **A complete σ_1 table.** We found no published table of exact first-order screening constants covering every NIST ground configuration for $N = 1-110$. **[VERIFY against the Z-expansion literature, e.g. Safronova et al. and Layzer et al. 1964, before claiming this.]**
2. **A blind-tested hybrid formula.** A single closed form joins exact σ_1 to a bounded fitted remainder. It is validated on all 5847 NIST successive IEs, with a pre-registered selection score and a blind extrapolation from $Z \leq 54$ to $Z \geq 55$. In the SHM literature we read, we found no blind out-of-range extrapolation test.
3. **A published SHM scored on every NIST row**, re-implemented and scored on all 5847 rows (§4.5).
4. **A teaching rule.** The 8-parameter pocket formula is a hand-calculable successor to Slater's rules: 4.68 % against 11.8 % on all rows, and 11.3 % against 11.6 % on blind S2.

Not claimed:

- a "first formula for all ionization energies", since SHMs and Dirac-Fock tables already cover all ions;
- a "first-principles formula", since only σ_1 and the one-electron corrections are first principles and the final model has 33 fitted parameters;
- priority over machine-learning work. We found no ML study of successive IEs of all atomic ions, but a systematic search is still required. **[TODO]**

The main lesson is methodological. Fitting alone interpolates to about 1.6 % and extrapolates catastrophically (170 %). Exact theory alone extrapolates smoothly but is about 70 % off overall. Fixing the Z^2 and Z terms by theory and confining the fit to a bounded $O(Z^0)$ remainder gives about 2 % inside the data and 6.6 % in blind extrapolation.

5.2 Where the formula fails

1. **Heavy near-neutral atoms.** The blind neutral first-IE MAPE is 21.8 %, and the all-data neutral MAPE is 7.5 % (6.4 % for u35). Valence screening in a neutral atom is an all-order, non-perturbative effect. The parameter-free LSDA Δ SCF does better on neutral atoms (3.3 % for $Z \leq 54$).
2. **An unbounded relativistic term.** The fitted r_c are all negative (-0.45 to -1.47), opposite to the direct relativistic contraction of s and $p^{1/2}$ electrons. The bracket is not bounded, which produces the negative L_r

value of the $Z \leq 54$ fit and the low superheavy 7p IEs. A physically bounded relativistic term is the obvious next step. It must be chosen on V1/V2/S1/S3 only.

3. **f-electron removal.** This is the least accurate class: 3.5 % on all data, and about 2.0–2.5 times too high (mean 2.2) for third IEs in blind S2.
4. **Multiplet structure and rearranged configurations.** The formula has one Hund term per removed electron and no term-dependent multiplet energies. Ions whose ground configuration rearranges on ionization are described by single-configuration inputs. This is also the origin of the 22 monotonicity violations.
5. **Parameter economy.** 19 of the 33 parameters are shrunk class deviations. The 9-parameter bounded variant (selection 2.51, blind S2 7.80 %, all data 2.90 %) is a reasonable lighter alternative. It was not the pre-registered winner.
6. **Reference data.** Most NIST values are labelled theoretical or semi-empirical (4617 + 919 of 5847). Agreement with them is partly agreement with other theory.

5.3 Why no exact closed form exists for $N \geq 2$

With $V = \sum 1/r_{ij}$, the many-electron Schrödinger equation does not separate. $E(Z)$ is analytic in $1/Z$ only up to a critical charge. For He, $1/Z_c \approx 1.0975$. Beyond first order, every coefficient E_k ($k \geq 2$) is an infinite sum over the hydrogenic continuum, with no known closed form even for two electrons; it is known only numerically [ScherrKnight1963]. A closed formula for all ions must therefore combine exact low orders with an approximation for the rest. The question is only which approximation, and how honestly it is validated. Our answer keeps the exact orders exact and confines the fit to the remainder, under a bound that keeps it from extrapolating wildly.

6. Conclusions

- **The formula.** The screened Rydberg formula gives every successive ionization energy of every element from the configuration alone, with 33 global parameters.
- **Accuracy.** 1.87 % MAPE (median 0.94 %) on 5847 NIST values; 7.5 % for neutral atoms; 0.0012 % for H-like ions.
- **Blind extrapolation.** A fit on $Z \leq 54$ predicts $Z = 55$ –110 with 6.60 % MAPE, under the 11.6 % of Slater's rules and the 13.6 % of a published parameter-free screened hydrogenic model. This holds because the large- Z behaviour is fixed exactly by first-order perturbation theory and the fitted remainder is bounded.
- **Remaining defects.** Heavy neutral atoms, f-electron removal and the sign of the fitted relativistic coefficients.
- **Most reusable result.** The table of exact rational first-order screening constants σ_1 for $N = 1$ –110.

Data and code availability

All code, data and results are in the project repository [TODO: archive with a DOI, e.g. Zenodo, and insert the link]:

- the NIST table (`data/nist_ie.csv`) and the shared scorer (`evaluate.py`);
- the exact first-order coefficients (`results/fp_zexp_coefficients.csv`, `results/fp_zexp_rows_coefficients.csv`);
- the final model and its parameters (`models/push_a/model.py`, `models/unified/final.py`, `results/uni_final_params.json`);
- predictions for all rows (`results/uni_predictions.csv`);

- the validation script (`models/unified/validate_blind.py`);
- the literature re-implementation (`models/literature/kregar_shm.py`).

A command-line and Python interface (`ionization.py`) evaluates the formula for any $Z \leq 118$ and $N \leq Z$.

AI-assistance disclosure

The computations, code, analysis and a draft of this text were produced with AI agents (Anthropic Claude) working under the author's direction. The author designed and directed the study and is responsible for its content. **Before submission, the author must:**

- independently verify every number against the results files;
- verify every reference against the original publication (several bibliographic fields are marked unverified in `docs/references.bib`);
- word this statement according to the target journal's policy.

An AI system is not listed as an author.

Acknowledgements

[TODO]

References

Only entries from `docs/references.bib` are used. Fields marked there as STANDARD, CITED-IN or UNVERIFIED must be checked against the publisher before submission. Most DOIs are omitted deliberately for that reason.

- [ClementiRaimondi1963] E. Clementi and D. L. Raimondi, Atomic screening constants from SCF functions, *J. Chem. Phys.* 38, 2686–2689 (1963).
- [ClementiRaimondiReinhardt1967] E. Clementi, D. L. Raimondi and W. P. Reinhardt, Atomic screening constants from SCF functions. II. Atoms with 37 to 86 electrons, *J. Chem. Phys.* 47, 1300–1307 (1967).
- [Chakravorty1993] S. J. Chakravorty, S. R. Gwaltney, E. R. Davidson, F. A. Parpia and C. Froese Fischer, Ground-state correlation energies for atomic ions with 3 to 18 electrons, *Phys. Rev. A* 47, 3649–3670 (1993).
- [Chung2005] H.-K. Chung, M. H. Chen, W. L. Morgan, Yu. Ralchenko and R. W. Lee, FLYCHK: generalized population kinetics and spectral model for rapid spectroscopic analysis for all elements, *High Energy Density Phys.* 1, 3–12 (2005).
- [Cowan1981] R. D. Cowan, *The Theory of Atomic Structure and Spectra* (University of California Press, Berkeley, 1981).
- [Crilly2023] A. J. Crilly et al., SpK: a fast atomic and microphysics code for the high-energy-density regime, *High Energy Density Phys.* (2023), doi:10.1016/j.hedp.2023.101053, arXiv:2211.16464.
- [DalgarnoStewart1958] A. Dalgarno and A. L. Stewart, A perturbation calculation of properties of the helium iso-electronic sequence, *Proc. R. Soc. Lond. A* 247, 245–259 (1958). [VERIFY]
- [DiRocco1992] H. O. Di Rocco, *Braz. J. Phys.* 22, 227 (1992). [VERIFY title]
- [DiRoccoLanzini2016] H. O. Di Rocco and F. Lanzini, Breit and quantum electrodynamics energy contributions in multielectron atoms from the relativistic screened hydrogenic model, *Braz. J. Phys.* 46, 175–183 (2016), doi:10.1007/s13538-015-0397-9.
- [Edlen1964] B. Edlén, Atomic spectra, in *Handbuch der Physik* vol. 27, ed. S. Flügge (Springer, Berlin, 1964), pp. 80–220. [VERIFY pages]

- [Faussurier1997] G. Faussurier, C. Blancard and A. Decoster, New screening coefficients for the hydrogenic ion model including l-splitting for fast calculations of atomic structure in plasmas, *J. Quant. Spectrosc. Radiat. Transfer* 58, 233 (1997). [VERIFY]
- [Faussurier2008] G. Faussurier, C. Blancard and P. Renaudin, Equation of state of dense plasmas using a screened-hydrogenic model with l-splitting, *High Energy Density Phys.* 4, 114–123 (2008).
- [FroeseFischer1997] C. Froese Fischer, T. Brage and P. Jönsson, *Computational Atomic Structure: An MCHF Approach* (Institute of Physics Publishing, Bristol, 1997).
- [KohnSham1965] W. Kohn and L. J. Sham, Self-consistent equations including exchange and correlation effects, *Phys. Rev.* 140, A1133–A1138 (1965).
- [Koopmans1934] T. Koopmans, Über die Zuordnung von Wellenfunktionen und Eigenwerten zu den einzelnen Elektronen eines Atoms, *Physica* 1, 104–113 (1934).
- [Kotochigova1997] S. Kotochigova, Z. H. Levine, E. L. Shirley, M. D. Stiles and C. W. Clark, Local-density-functional calculations of the energy of atoms, *Phys. Rev. A* 55, 191–199 (1997).
- [Kregar1984] M. Kregar, *Phys. Scr.* 29, 438 (1984). [VERIFY title]
- [Kregar1985] M. Kregar, *Phys. Scr.* 31, 246 (1985). [VERIFY title]
- [LanziniDiRocco2015] F. Lanzini and H. O. Di Rocco, Screening parameters for the relativistic hydrogenic model, *High Energy Density Phys.* 17, 240–247 (2015).
- [Layzer1959] D. Layzer, On a screening theory of atomic spectra, *Ann. Phys. (N.Y.)* 8, 271 (1959). [VERIFY pages]
- [LayzerEtAl1964] D. Layzer et al., *Ann. Phys. (N.Y.)* 29, 101 (1964). [VERIFY title and authors]
- [Layzer1967] D. Layzer, *Int. J. Quantum Chem.* 1S, 45 (1967). [VERIFY title]
- [Martel1998] P. Martel, J. G. Rubiano, J. M. Gil, L. Doreste and E. Mínguez, *J. Quant. Spectrosc. Radiat. Transfer* 60, 623 (1998). [VERIFY title]
- [Mayer1947] H. Mayer, Methods of opacity calculations, Los Alamos Scientific Laboratory report LA-647 (1947).
- [Mendoza2011] M. A. Mendoza, J. G. Rubiano, J. M. Gil, R. Rodríguez, R. Florido, P. Martel and E. Mínguez, A new set of relativistic screening constants for the screened hydrogenic model, *High Energy Density Phys.* 7, 169–179 (2011). [VERIFY volume/pages]
- [More1982] R. M. More, Electronic energy levels in dense plasmas, *J. Quant. Spectrosc. Radiat. Transfer* 27, 345–357 (1982).
- [NISTASD] A. Kramida, Yu. Ralchenko, J. Reader and NIST ASD Team, NIST Atomic Spectra Database (ver. 5.x), National Institute of Standards and Technology, Gaithersburg, MD, <https://physics.nist.gov/asd>. [Insert exact version and access date]
- [Pomarico2005] J. Pomarico, D. I. Iriarte and H. O. Di Rocco, An efficient screening approach to be used in plasma modeling and ion-surface collision experiments, *Braz. J. Phys.* 35(1), 130–135 (2005).
- [Rodrigues2004] G. C. Rodrigues, P. Indelicato, J. P. Santos, P. Patté and F. Parente, Systematic calculation of total atomic energies of ground state configurations, *At. Data Nucl. Data Tables* 86, 117–233 (2004), doi:10.1016/j.adt.2003.11.005.
- [Rozsnyai1972] B. F. Rozsnyai, Relativistic Hartree-Fock-Slater calculations for arbitrary temperature and matter density, *Phys. Rev. A* 5, 1137–1149 (1972). [VERIFY pages]
- [Rubiano2002] J. G. Rubiano, R. Rodríguez, J. M. Gil, F. H. Ruano, P. Martel and E. Mínguez, A screened hydrogenic model using analytical potentials, *J. Quant. Spectrosc. Radiat. Transfer* 72, 575 (2002).
- [Safronova1993] U. I. Safronova et al., *Phys. Scr.* 47, 364 (1993). [VERIFY title and authors]
- [ScherrKnight1963] C. W. Scherr and R. E. Knight, Two-electron atoms III. A sixth-order perturbation study of the 1^1S ground state, *Rev. Mod. Phys.* 35, 436–442 (1963). [VERIFY pages]
- [Slater1930] J. C. Slater, Atomic shielding constants, *Phys. Rev.* 36, 57–64 (1930).

- [YerokhinShabaev2015] V. A. Yerokhin and V. M. Shabaev, Lamb shift of $n = 1$ and $n = 2$ states of hydrogen-like atoms, $1 \leq Z \leq 110$, J. Phys. Chem. Ref. Data 44, 033103 (2015).

Appendix A. Fitted parameters (all-data fit)

Table A1. Source: `results/uni_final_params.json`, which is identical to `results/pa_params.json`.

block	values
τ_g (same, in, core, df, out)	0.3928, 0.7879, 2.2243, 2.5488, 10.5688
κ	1.7527 (1.8027 as used in the code, which applies $ \kappa + 0.05$)
$\delta\tau_c$ (19 classes)	same_s -0.1552, same_p 0.0098, same_d 0.0004, same_f 0.1457, sn_in_p -0.1133, n1_sp_sp -0.3916, n1_sp_d 0.3009, n1_sp_f 0.2042, n2_sp_sp -0.2195, n2_sp_df -0.0622, d_near -0.6962, n1_d_d -0.2689, n1_d_f 0.3286, f_near 0.5476, n1_f_f -0.0918, out_d -0.0176, out_f 0.0173, core_sp 0.2828, core_df 0.1812
r_c (s, $p^{1/2}$, $p^{3/2}$, d, f)	-0.4543, -0.8129, -1.1533, -0.8426, -1.4672
x_l (p, d, f)	0.2049, 0.4402, 0.4612

Pocket formula (8 parameters, all-data fit, `results/uni_params.json`).

$$Z_{\text{eff}} = Z - \sum_g s_g \nu_g - t \frac{N - 1}{Z - N + 1 + \kappa}, \quad \text{IE} = \text{Ry} \frac{Z_{\text{eff}}^2}{n^2} \left[1 + \frac{(Z_{\text{eff}} \alpha)^2}{n^2} \left(\frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) \right] + \text{Ry} \frac{x K_l(k)}{n^2}$$

The fitted values are s_same 0.7499, s_in 0.7514, s_core 0.8432, s_df 0.8408, s_out 1.0497, t 1.3725, κ 9.815 (9.865 as used) and x 0.508.